ICAD RT-5 · Set 5 — Answer Sheet

Exam 16-11-2026 · 75 Q · 300 marks · 3 hours · +4 correct, −1 wrong, 0 blank

Name ..... Roll No ..... Date .....

Physics

1	a	b	c	d
2	a	b	c	d
3	a	b	c	d
4	a	b	c	d
5	a	b	c	d
6	a	b	c	d
7	a	b	c	d
8	a	b	c	d
9	a	b	c	d
10	a	b	c	d
11	a	b	c	d
12	a	b	c	d
13	a	b	c	d
14	a	b	c	d
15	a	b	c	d
16	a	b	c	d
17	a	b	c	d
18	a	b	c	d
19	a	b	c	d
20	a	b	c	d

Numerical — write the value

21	
22	
23	
24	
25	

Chemistry

1	a	b	c	d
2	a	b	c	d
3	a	b	c	d
4	a	b	c	d
5	a	b	c	d
6	a	b	c	d
7	a	b	c	d
8	a	b	c	d
9	a	b	c	d
10	a	b	c	d
11	a	b	c	d
12	a	b	c	d
13	a	b	c	d
14	a	b	c	d
15	a	b	c	d
16	a	b	c	d
17	a	b	c	d
18	a	b	c	d
19	a	b	c	d
20	a	b	c	d

Numerical — write the value

21	
22	
23	
24	
25	

Mathematics

1	a	b	c	d
2	a	b	c	d
3	a	b	c	d
4	a	b	c	d
5	a	b	c	d
6	a	b	c	d
7	a	b	c	d
8	a	b	c	d
9	a	b	c	d
10	a	b	c	d
11	a	b	c	d
12	a	b	c	d
13	a	b	c	d
14	a	b	c	d
15	a	b	c	d
16	a	b	c	d
17	a	b	c	d
18	a	b	c	d
19	a	b	c	d
20	a	b	c	d

Numerical — write the value

21	
22	
23	
24	
25	

Bubbles — single correct · Box — write the value · each subject out of 100

ICAD RT-5 · Set 5

Exam 16-11-2026 · 25 Physics · 25 Chemistry · 25 Mathematics · 300 marks · 3 hours · +4 / −1

Physics

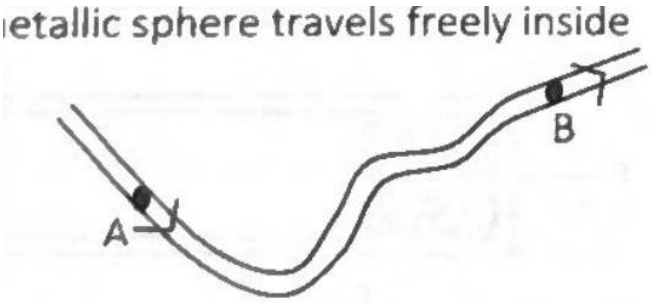
PART A · Q1–20 · SINGLE CORRECT

1. P4L2VSuggQ4 A car starts from rest and moves with constant acceleration. Ratio of the distance covered in the nth second to that covered in n seconds: (a) $2/n^2 - 1/n$ (b) $2/n^2 + 1/n$ (c) $2/n - 1/n^2$ (d) $2/n + 1/n^2$

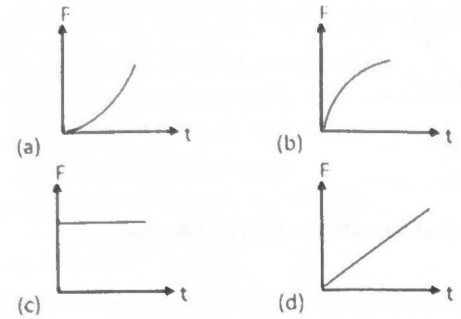
2. P4L10V5Q9 Two particles A and B are placed in gravity free space at (0, 0, 0) m and (30, 0, 0) m respectively. Particle A is projected with a velocity $(5\hat{i} + 10\hat{j} + 5\hat{k})$ ms^{-1}, while particle B is projected with a velocity $(10\hat{i} + 5\hat{j} + 5\hat{k})$ ms^{-1} simultaneously. Then: (a) they will collide at (10, 20, 10) m (b) they will collide at (10, 10, 10) m (c) they will never collide (d) they will collide at 2 second

3. P4XVCAT3Q11 Two tall buildings are 30 m apart. The speed with which a ball must be thrown horizontally from a window 150 m above the ground in one building so that it enters a window 27.5 m from the ground in the other building is: (take g = 9.8 m/s^2) (a) 2 ms^{-1} (b) 6 ms^{-1} (c) 4 ms^{-1} (d) 8 ms^{-1}

4. P6L1V5Q5 A small metallic sphere travels freely inside a tunnel as shown (from A to B). V_A = $(4\hat{i} + 3\hat{j})$ m/s and the average force that acted upon the sphere as applied by the tunnel equals f_avg = $(4\hat{i} + 3\hat{j})$ N; then the velocity of the sphere at B equals [mass of sphere = 2 kg] [time taken to travel the curved path = 4 s]. (Treat the tunnel force as the only force on the sphere — the tunnel lies in a horizontal plane.) (a) V_B = $(9\hat{i} + 12\hat{j})$ m/s (b) V_B = $(9\hat{i} - 12\hat{j})$ m/s (c) V_B = $(12\hat{i} - 9\hat{j})$ m/s (d) V_B = $(12\hat{i} + 9\hat{j})$ m/s

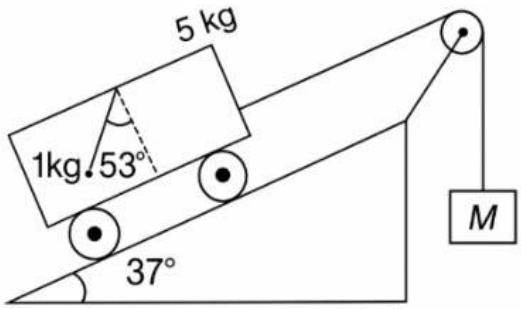


5. P6L1V6Q2 Linear momentum for an object varies as $p = 3t^3 + 4t^2$. The force vs time curve for the object is (a) [F–t graph: curve from the origin, concave up] (b) [curve from the origin, concave down] (c) [horizontal line, F constant] (d) [straight line through the origin]

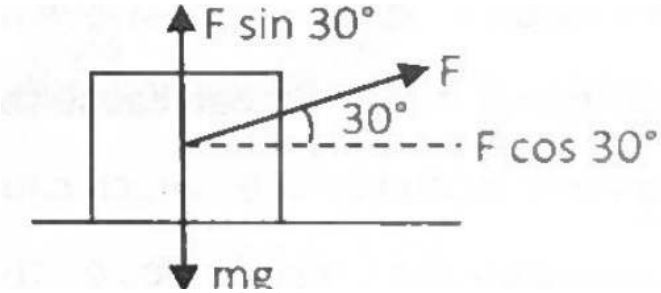


6. P6L2V1Q3 Which type of force causes the change in position (state of motion) of a body? (a) Unbalanced Force (b) Balanced Force (c) More than one of the above (d) None of the above

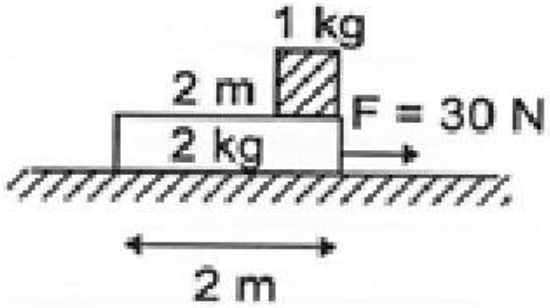
7. P6XVCAT4Q16 Find mass (M) of the hanging block, if the string (of the 1 kg bob hanging inside the 5 kg trolley on the 37° incline, trolley pulled up the incline by a string over a pulley to M) in steady state makes an angle of 53° with normal to the trolley as shown in figure. (Smooth incline, g = 10 m/s^2.) (a) 8 kg (b) 10 kg (c) 12 kg (d) 16 kg



8. P7L1VSuggQ5 A block is pulled to the right at constant velocity by a force of 10 N acting 30° above the horizontal. If the coefficient of sliding friction between the block and the surface is 0.5, the weight of the block is: (a) 20 N (b) 22.32 N (c) 56 N (d) 72.4 N.



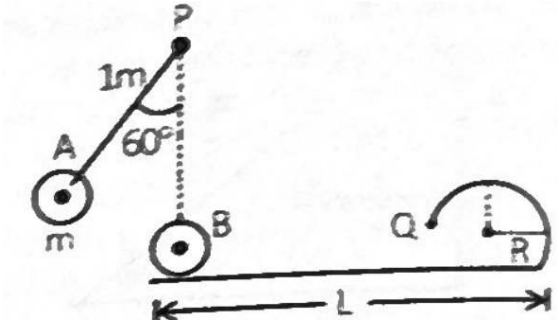
9. P7XVRT4Q19 A block of mass 1 kg is placed over a plank of mass 2 kg. The length of the plank is 2 m. Coefficient of friction between the block and the plank is 0.5 and the ground over which plank is frictionless. A constant force F = 30 N is applied on the plank in horizontal direction. The time after which the block will separate from the plank is (g = 10 m/s^2) (a) 0.73 s (b) 1.2 s (c) 0.62 s (d) 1.6 s



10. P8L4V3Q3 A bullet when fired at a target has its velocity decreased to 50 % after penetrating 30 cm into it. Then the additional thickness it will penetrate in cm before coming to rest is: (a) 10 cm (b) 30 cm (c) 40 cm (d) 60 cm

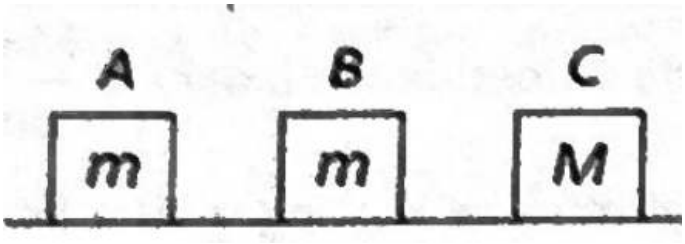
11. P8L5V3Q3 The potential energy φ in joule of a particle of mass 1 kg moving in the x-y plane obeys the law φ = 3x + 4y. Here, x and y are in metres. If the particle is at rest at (6 m, 8 m) at time t = 0, then the work done by the conservative force on the particle from the initial position to the instant when it crosses the x-axis is (a) 25 J (b) −25 J (c) 50 J (d) −50 J

12. P8SBVPYQQ1 A small bob A of mass m is attached to a massless rigid rod of length 1 m pivoted at point P and kept at an angle of 60° with the vertical. At a distance of 1 m below point P, an identical bob B is kept at rest on a smooth horizontal surface that extends to a circular track of radius R. If bob B just manages to complete the circular path of radius R up to a point Q after being hit elastically by bob A, then radius R is _____ m — (a) 3/5 (b) 1/5 (c) (2+√3)/5 (d) (2−√3)/5 — (PYQ)



13. P9L4V5Q5 A thin uniform rod of length l and mass m is swinging freely about a horizontal axis passing through its end. Its maximum angular speed is ω. Its centre of mass rises to a maximum height of (a) $1/3 \cdot l^2\omega^2/g$ (b) $1/6 \cdot l\omega/g$ (c) $1/2 \cdot l^2\omega^2/g$ (d) $1/6 \cdot l^2\omega^2/g$

14. P9L4V5Q9 Three blocks A, B and C are lying on a smooth horizontal surface as shown in the figure. A and B have equal masses m while C has mass M. Block A is given an initial speed v towards B due to which it collides with B perfectly inelastically. The combined mass collides with C, also perfectly inelastically. 5/6th of the initial kinetic energy is lost in whole process. What is value of M/m? (a) 4 (b) 2 (c) 3 (d) 5



PART B · Q21–25 · NUMERICAL VALUE

21. P2XVRT2Q22 The motion of a body is given by the equation $dv(t)/dt = 6 - 3v(t)$, where $v(t)$ at time t is in m s −1 and t is in seconds. If the body was at rest at t = 0, the terminal speed (in m s −1) is

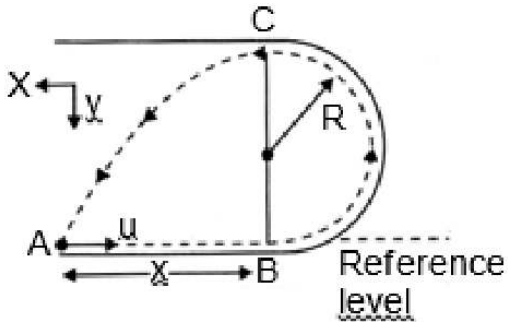
22. P3SBVPYQQ7 A force $-P\hat{k}$ acts at the origin; the torque about the point (2, −3) is $P(a\hat{i} + b\hat{j})$. If a/b = x/2 , find x

23. P4XVCAT3Q21 A bird flies for 4 s with a velocity v = (t − 2) m/s in a straight line, where t = time in second. Calculate the distance covered by the bird in meter.

15. P10L1V6Q6 The magnitude of displacement of a particle moving in a circle of radius a with constant angular speed ω varies with time t as: (a) 2a sinωt (b) 2a sin(ωt/2) (c) 2a cosωt (d) 2a cos(ωt/2)

16. P10L4V6Q10 Multiple Correct Answer Question. A small sphere of mass m suspended by a thread is first taken aside so that the thread forms the right angle with the vertical and then released, then: (one or more options may be correct) (a) total acceleration of sphere as a function of θ is $g\sqrt{(1 + 3\cos^2\theta)}$ (b) thread tension as a function of θ is T = 3mg cosθ (c) the angle θ between the thread and the vertical at the moment when the total acceleration vector of the sphere is directed horizontally is cos −1 (1/√3) (d) thread tension at the moment when the vertical component of the sphere's velocity is maximum will be mg

17. P10XVCAT6Q19 A small ball is rolled with speed u from point A along a smooth circular track as shown in fig., then what is the minimum value of x for which the ball can reach the point of projection after reaching C? (a) 2R (b) 5R (c) 3R (d) (5/2)R

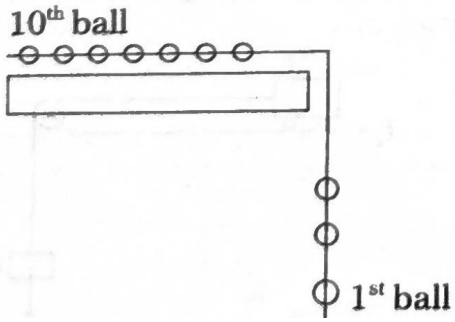


18. P11L2V5Q1 A thin uniform wire is bent to form the two equal sides AB and AC of triangle ABC, where AB = AC = 5 cm. The third side BC of length 6 cm is made from uniform wire of twice the density (mass per unit length) of the first. The distance of centre of mass from A is (a) 34/11 cm (b) 11/34 cm (c) 34/9 cm (d) 11/45 cm

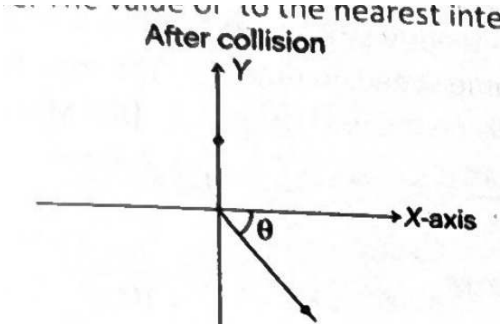
19. P11L3V1Q4 In the formula for the center of mass along the x -axis, what does the numerator represent? Options: (a) Total mass (b) Area-weighted sum of x -coordinates (c) Moment of inertia (d) Volume

20. P11L4V2Q1 Two persons of masses 55 kg and 65 kg respectively, are at the opposite ends of a boat. The length of the boat is 3.0 m and weighs 100 kg. The 55 kg man walks up to the 65 kg man and sits with him. If the boat is in still water the centre of mass of the system shifts by (a) 3.0 m (b) 2.3 m (c) Zero (d) 0.75 m

24. P6SBVPYQQ7 A system of 10 balls each of mass 2 kg are connected via massless and unstretchable string. The system is allowed to slip over the edge of a smooth table as shown in figure. Tension on the string between the 7th and 8th ball is _____ N when the 6th ball just leaves the table. ($g = 10\ m/s^2$)



25. P9L5L6V6Q2 A ball of mass 10 kg moving with a velocity $10\sqrt{3}$ ms^{-1} along X-axis, hits another ball of mass 20 kg, which is at rest. After collision, the first ball comes to rest and the second one disintegrates into two equal pieces. One of the pieces starts moving along Y-axis at a speed of 10 m/s. The second piece starts moving at a speed of 20 m/s at an angle θ (degree) with respect to the X-axis. The value of θ to the nearest integer is _____.



Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C1L7VSuggQ3 For two gases A and B at the same temperature, if M_A = 4M_B, what is the ratio of their most-probable speeds u_mp,A : u_mp,B? (a) 1 : 2 (b) 2 : 1 (c) 1 : 4 (d) 4 : 1

2. C2L1VSuggQ3 If the mass of the nucleus accounts for almost the entire atomic mass, what does this indicate about the distribution of mass and charge within an atom? (a) Both mass and charge are uniformly distributed. (b) Mass is concentrated in the nucleus, and charge is spread over the entire atom. (c) Mass and charge are equally shared by electrons and protons. (d) All mass and charge are carried only by electrons.

3. C2L9V5Q2 Which of the following set of quantum numbers belong to highest energy? (a) n = 4, ℓ = 0, m = 0, s = +1/2 (b) n = 3, ℓ = 0, m = 0, s = +1/2 (c) n = 3, ℓ = 1, m = 1, s = +1/2 (d) n = 3, ℓ = 2, m = 1, s = +1/2

4. C2XVRT4Q31 Which of the following sets of quantum numbers represents the highest energy orbital (in a multi-electron atom)? (a) n = 3, l = 0 (b) n = 3, l = 1 (c) n = 4, l = 0 (d) n = 3, l = 2

5. C3L1V6Q6 If we consider that 1/6th in place of 1/12th mass of carbon atom is taken to be the relative atomic mass unit, the mass of one mole of a substance will: (a) decrease twice (b) increase two fold (c) remains unchanged (d) be a function of the molecular mass of the substance

6. C3L10V5Q7 The molality of 1 L solution of H_2SO_4 (M = 98 g mol^{-1}) with x% (W/W) is equal to 9. The weight of the solvent present in the solution is 910 g. The value of x is: (a) 90 (b) 46.86 (c) 40.13 (d) 9

7. C4L3V4Q10 Which of the following atom or ions has the smallest size? (a) F (b) F^- (c) O (d) N

8. C4L4V5Q6 Select the correct alternate about (IE): (a) IE of Be^+ > IE of B; (b) IE of O > IE of N; (c) Both (a) and (b); (d) None of these.

9. C4SBVPYQQ12 Given below are two statements: Statement (I): The first ionization energy of Pb is greater than that of Sn. Statement (II): The first ionization energy of Ge is greater than that of Si. In the light of the above statements, choose the correct answer from the options given below: (a) Statement I is false but Statement II is true (b) Statement I is true but Statement II is false (c) Both Statement I and Statement II are true (d) Both Statement I and Statement II are false

10. C5L1V5Q7 In a metallic crystal the (a) Valence electrons remain within the fields of influence of their own kernels. (b) Valence electrons constitute a sea of mobile electrons. (c) Valence electrons are localised between the two kernels (d) Kernels as well as the electrons move rapidly.

11. C5L4V3Q5 For XeF_2 , 2 bond pairs and 3 lone pairs exist. What is hybridisation and shape? (a) sp^3d, linear (b) sp^3d^2, octahedral (c) sp^3, bent (d) sp^2, trigonal planar

12. C5L6V3Q3 In H_2O, what hybridisation leads to the bond angle 104.5°? (a) sp^3 (b) sp^2 (c) sp (d) dsp^2

13. C5XVRT4Q42 Choose the incorrect order: (a) $N_2 > N_2^+ = N_2^- > N_2^{2+}$ (Bond order) (b) $N_2 > N_2^+ = N_2^- < N_2^{2+}$ (Bond dissociation energy) (c) $O_2^{2+} > O_2^+ > O_2^- > O_2^{2-}$ (Bond order) (d) $O_2 < O$ (Ionisation energy)

14. C6L1V5Q7 Select the correct option based on the statements given below: Statement I. Specific heat is an intensive property. Statement II. Heat capacity is an extensive property. (a) Statement I is correct; Statement II is correct (b) Statement I is correct; Statement II is incorrect (c) Statement I is incorrect; Statement II is correct (d) Statement I is incorrect; Statement II is incorrect

15. C6L4VSuggQ2 For a process carried out at constant volume, the heat exchanged by the system is equal to: (a) ΔH (b) −PΔV (c) ΔU (d) ΔU + PΔV

16. C6L6V3Q1 A substance receives 500 J of heat, raising the temperature of 2 kg by 10 K. Find specific heat capacity and then convert it to J/g·K. (a) 0.25 J/g·K (b) 0.025 J/g·K (c) 2.5 J/g·K (d) 0.5 J/g·K

17. C6L10V4Q6 ΔS° will be highest for the reaction: (a) Ca(s) + (1/2)O 2 (g) → CaO(s) (b) CaCO 3 (s) → CaO(s) + CO 2 (g) (c) C(s) + O 2 (g) → CO 2 (g) (d) N 2 (g) + O 2 (g) → 2NO(g)

18. C6L10VSuggQ5 A process at constant temperature absorbs heat reversibly from surroundings. Which condition must be satisfied for the process to be spontaneous? (a) ΔS system < 0 (b) ΔS surroundings > 0 (c) ΔS universe > 0 (d) ΔS universe = 0

19. C6L12V5Q1 In the conversion of limestone to lime: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$. The values of ΔH° and ΔS° are 179.1 kJ mol^{-1} and 160.2 J K^{-1} mol^{-1} respectively at 298 K and 1 bar. Assuming ΔH° and ΔS° remain constant with temperature, above which temperature will the conversion of limestone to lime be spontaneous? (a) 1118 K (b) 1008 K (c) 1200 K (d) 845 K

20. C6SBVPYQQ26 500 J of energy is transferred as heat to 0.5 mol of Argon gas (ideal, monoatomic) at 298 K and a constant pressure of 1.00 atm. The final temperature and the change in internal energy respectively are: (Given: R = 8.3 J K^{-1} mol^{-1}) (a) 378 K and 500 J (b) 368 K and 500 J (c) 348 K and 300 J (d) 378 K and 300 J

PART B · Q21–25 · NUMERICAL VALUE

21. C1XVRT4Q35 50 mL of a gas A diffuse through a membrane in the same time as for the diffusion of 40 mL of gas B under identical conditions of pressure and temperature. If the molecular mass of A is 64, that of B would be: (A) 100 (B) 250 (C) 200 (D) 80

22. C3XVRT2Q26 Excess F 2 (g) reacts at 150°C and 1.0 atm pressure with Br 2 (g) to give a compound BrF n . If 423 mL of Br 2 (g) at the same temperature and pressure produced 4.2 g of BrF n , what is n? (Br = 80, F = 19) (A) 3 (B) 1 (C) 5 (D) 7

23. C4XVCAT4Q49 Energy (kJ) required to convert 7.974 g of gaseous Cs atoms to Cs^+ ions. IE_1(Cs) = 374 kJ/mol, atomic mass = 132.9 amu. (round to nearest integer)

24. C5XVCAT5Q49 Assuming there is no 2s – 2p mixing, the number of paramagnetic species amongst the following: H_2, He_2^+, Li_2, Be_2, B_2, C_2, C_2^{2-}, N_2^+, N_2, O_2, O_2^-, S_2 & F_2 are _____

25. C6SBVPYQQ47 The enthalpy change for the conversion of (1/2)Cl 2 (g) to Cl – (aq) is (–)_____ kJ mol –1 . (Nearest integer) Given Δ dis H°(Cl 2) = 240 kJ mol –1 , Δ eg H°(Cl) = –350 kJ mol –1 , Δ hyd H°(Cl –) = –380 kJ mol –1 .

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M2L1V5Q5 If a set A has 3 elements and B has 6 elements, then the minimum number of elements in A ∪ B is (a) 6 (b) 3 (c) 9 (d) None of these

2. M4L1VSuggQ3 Find the domain of the function f(x) = (2x + 3)/(x – 1). (a) R\{1} (b) R\{0} (c) R\{–3} (d) R

3. M4XVRT4Q53 If $f(x) = x^n + 1$ and f(4) = 65, then the value of f(6) is : (a) 220 (b) 217 (c) 214 (d) 37

4. M5AL2VSuggQ3 If tan α = m/(m + 1) and tan β = 1/(2m + 1), where m > 0 and α, β are acute, then α + β = (a) π/3 (b) π/2 (c) π/4 (d) π/6

5. M6L4V5Q6 Sum of all real roots of the equation $|x – 3|^2 + 5|x – 2| + 6 = 0$ (a) 0 (b) 1 (c) 2 (d) No real roots

6. M6SBVPYQQ1 Let f be a function such that 3f(x) + 2f(m/(19x)) = 5x, x ≠ 0, where m = Σ (from i=1 to 9) i^2. Then f(5) – f(2) is equal to (a) –9 (b) 9 (c) 18 (d) 36

7. M7L3V5Q5 If b > 1, x > 0 and (2x)^(log_b 2) – (3x)^(log_b 3) = 0, then x is: (a) 1/216 (b) 1/6 (c) 1 (d) 6

8. M7L4V5Q3 If log_x{log4(log_x(5x² + 4x³))} = 0, then x = (a) 2 (b) 3 (c) 4 (d) 5

9. M7XVRT4Q52 $\log_2[\log_4(\log_{10} 16^4 + \log_{10} 25^8)]$ simplifies to : (a) an irrational (b) an odd prime (c) a composite (d) unity

10. M8AL1V4Q10 The 11 th term of the A.P. –5, –5/2, 0, 5/2, ... is (NCERT) (a) –20 (b) 20 (c) –30 (d) 30

11. M8AL1V6Q6 Let T_r be the r th term of an A.P. for r = 1, 2, ... If for some positive integers m and n we have $T_m = 1/n$ and $T_n = 1/m$, then T_{mn} = (a) –1/mn (b) 1/m + 1/n (c) 1 (d) 0

12. M8BL3VSuggQ5 If $S_n = 1 + 1/2 + 1/2^2 + ... + 1/2^{(n–1)}$, then the least value of n such that $2 – S_n < 1/100$ is (a) 7 (b) 8 (c) 9 (d) 10

13. M8BXVRT4Q58 The value of (0.2)^(log_√5 (1/4 + 1/8 + 1/16 + ...)) is (a) 1 (b) 2 (c) 1/2 (d) 4

14. M8DL2V2Q2 Find T 10 of the sequence 3, 10, 21, 36, ... (a) 208 (b) 209 (c) 210 (d) 215

15. M8DL3V2Q2 Evaluate S = 1/(3 × 4). (a) 1/7 (b) 1/10 (c) 1/12 (d) 1/15

16. M9L3V3Q4 The conjugate of z = 3 + 4i is (a) The reflection in the x-axis (b) The reflection in the y-axis (c) The reflection in the origin (d) The reflection of -3 + 4i in the real axis

17. M9L4V5Q9 For all complex numbers z_1, z_2 satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$, the minimum value of $|z_1 - z_2|$ is (a) 0 (b) 2 (c) 7 (d) 17

18. M9L5V5Q3 If |z| = 6, arg(z) = 7π/6 then z (a) -3√3 + 3i (b) 3√2 + 3i (c) 3√3 - 3i (d) -3√3 - 3i

19. M9L6V5Q6 If $z = 1/(1 – \cos\theta – i\sin\theta)$, then Re(z) = (a) 0 (b) 1/2 (c) cot(θ/2) (d) (1/2)cot(θ/2)

20. M9L7V4Q10 If ω is a non-real (complex) cube root of unity then $(1 – \omega + \omega^2)^5 + (1 + \omega – \omega^2)^5$ = (a) 32 (b) 31 (c) 30 (d) 28

PART B · Q21–25 · NUMERICAL VALUE

21. M2SBVPYQQ5 The sum of all the elements of the set {α ∈ {1, 2, ... 100} : HCF(α, 24) = 1} is _____

22. M3XVRT1Q74 If $(\cos^2 48° – \sin^2 12°)/(\sin^2 24° – \sin^2 6°) = (\alpha + \beta\sqrt{5})/2$, where α, β ∈ N, then α + β is ___.

23. M5CXVRT2Q72 If $A + B + C = \pi$, then the value of $\cos^2 A + \cos^2 B + \cos^2 C + 2 \cos A \cos B \cos C$ is equal to

24. M8AXVCAT5Q72 Number of identical terms in the sequence 2, 5, 8, 11, ... upto 100 terms and 3, 5, 7, 9, 11, ... upto 100 terms are

25. M8DSBVPYQQ6 The series of positive multiples of 3 is divided into sets: {3}, {6, 9, 12}, {15, 18, 21, 24, 27}, ... Then the sum of the elements in the 11 th set is equal to _____